THE SPACE OF A DECENT STACK WITH QUASI-COMPACT INERTIA IS SOBER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The reduced induced closed substack structure on an irreducible closed subset. Invoked inside the counted proof (source lines 10584-10587) as a cross-chapter pointer. Turns the given irreducible closed subset into a stack to which the induction hypothesis applies.
- The permanence of decency under representable morphisms and the base change description of the inertia. Invoked inside the counted proof (source lines 10588-10597). Shows that the reduced closed substack again satisfies the hypotheses.
- The existence of a dense open substack that is a gerbe. Invoked inside the counted proof (source lines 10602-10604). Reduces further to the case of a gerbe.
- The bijection of the space of a gerbe with the space of its base and the sobriety of decent algebraic spaces. Invoked near the end of the counted proof (source lines 10605-10616). Identifies the space of the gerbe with a decent algebraic space and applies sobriety there.

Source reference. Original reference decent-spaces-proposition-reasonable-sober: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10616–10616. Navigation pointer only: the counted text refers to the Stacks item decent-spaces-proposition-reasonable-sober, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-base-change-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1984–1984. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gerbe-bijection-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6618–6618. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-bijection-points, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gerbe-decent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10493–10493. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-decent, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-kolmogorov: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10519–10519. Navigation pointer only: the counted text refers to the Stacks item lemma-kolmogorov, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-monomorphism-cartesian-square-inertia: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1480–1480. Navigation pointer only: the counted text refers to the Stacks item lemma-monomorphism-cartesian-square-inertia, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-representable-decent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10425–10425. Navigation pointer only: the counted text refers to the Stacks item lemma-representable-decent, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original proposition proposition-open-stratum: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6746–6746. Navigation pointer only: the counted text refers to the Stacks item proposition-open-stratum, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-definition-reduced-induced-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6968–6968. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-definition-reduced-induced-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Proposition 0.1. *Let $\mathcal{X}$ be a decent algebraic stack such that $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is quasi-compact. Then $|\mathcal{X}|$ is sober.*

Proof. By Lemma lemma-kolmogorov we know that $|\mathcal{X}|$ is Kolmogorov (in fact we will reprove this). Let $T \subset |\mathcal{X}|$ be an irreducible closed subset. We have to show T has a generic point. Let $\mathcal{Z} \subset \mathcal{X}$ be the reduced induced closed substack corresponding to T, see Properties of Stacks, Definition stacks-properties-definition-reduced-induced-stack. Since $\mathcal{Z} \to \mathcal{X}$ is a closed immersion, we see that $\mathcal{Z}$ is a decent algebraic stack, see Lemma lemma-representable-decent. Also, the morphism $\mathcal{I}_\mathcal{Z} \to \mathcal{Z}$ is the base change of $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ (Lemma lemma-monomorphism-cartesian-square-inertia). Hence $\mathcal{I}_\mathcal{Z} \to \mathcal{Z}$ is quasi-compact (Lemma lemma-base-change-quasi-compact). Thus we reduce to the case discussed in the next paragraph.

Assume $\mathcal{X}$ is decent, $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is quasi-compact, $\mathcal{X}$ is reduced, and $|\mathcal{X}|$ irreducible. We have to show $|\mathcal{X}|$ has a generic point. By Proposition proposition-open-stratum. there exists a dense open substack $\mathcal{U} \subset \mathcal{X}$ which is a gerbe. In other words, $|\mathcal{U}| \subset |\mathcal{X}|$ is open dense. Thus we may assume that $\mathcal{X}$ is a gerbe in addition to all the other properties. Say $\mathcal{X} \to X$ turns $\mathcal{X}$ into a gerbe over the algebraic space X. Then $|\mathcal{X}| \cong |X|$ by Lemma lemma-gerbe-bijection-points. In particular, X is quasi-compact and $|X|$ is irreducible. Also, by Lemma lemma-gerbe-decent we see that X is a decent algebraic space. Then $|\mathcal{X}| = |X|$ is sober by Decent Spaces, Proposition decent-spaces-proposition-reasonable-sober and hence has a (unique) generic point. □